IMAT 모의고사 5회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Placebo effects are often dismissed as "imaginary", but brain-imaging studies show that expecting pain relief can trigger the release of the body's own opioids. Remarkably, some patients improve even when told openly that they are receiving a placebo, although such "open-label" trials have mostly involved conditions with subjective symptoms, such as chronic back pain or irritable bowel syndrome. Researchers caution that a placebo cannot shrink a tumour or cure an infection, and that relying on it in place of effective treatment would be unethical.

1. The author mentions brain-imaging studies primarily in order to:

A. argue that pain is imaginary.
B. criticise open-label trials.
C. show that placebo responses have a measurable biological basis.
D. prove that placebos can cure infections.
E. recommend placebos instead of drugs.

2. Which of the following is an assumption underlying the researchers' caution in the last sentence?

A. Open-label placebos are illegal.
B. Effective treatments exist for some conditions that placebos cannot treat.
C. Tumours respond only to placebos.
D. Placebos never have any effect.
E. All symptoms are subjective.

3. Which scientist discovered penicillin in 1928?

A. Jonas Salk
B. Selman Waksman
C. Paul Ehrlich
D. Joseph Lister
E. Alexander Fleming

4. Which of the following countries is NOT a member of the European Union (as of 2026)?

A. Portugal
B. Norway
C. Austria
D. Ireland
E. Croatia

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "Every time it rains, the clinic car park floods. The car park is flooded today." Which of the following is true?

A. The flooding may have another cause; we cannot be sure it rained.
B. Rain never causes flooding.
C. It did not rain today.
D. It must have rained today.
E. It will rain tomorrow.

6. A patient's weight dropped from 80 kg to 68 kg. By what percentage did it decrease?

A. 12%
B. 17.6%
C. 18%
D. 15%
E. 20%

7. Four friends — Luca, Maria, Nina and Omar — each study a different subject: anatomy, biology, chemistry, physics. Luca does not study biology or physics. Maria studies chemistry. Nina does not study physics. What does Omar study?

A. Physics
B. Chemistry
C. Anatomy
D. Biology
E. Cannot be determined

8. A cube of side 3 cm is painted on all faces and then cut into 1 cm cubes. How many small cubes have exactly two painted faces?

A. 18
B. 6
C. 8
D. 27
E. 12

9. A dose of 5 mg/kg of a drug is prescribed. The drug is available as a solution of 25 mg/mL. What volume should be given to a child weighing 20 kg?

A. 0.8 mL
B. 5 mL
C. 2 mL
D. 4 mL
E. 100 mL

Section 3 — Biology (Q10–32)

10. Which of the following best explains why mitochondria are thought to have evolved from free-living bacteria?

A. They carry out photosynthesis.
B. They contain cellulose.
C. They lack membranes.
D. They are found in all prokaryotes.
E. They have their own circular DNA and 70S-like ribosomes, and a double membrane.

11. A DNA molecule replicates twice in a medium containing only ^{14}N, having originally been fully labelled with ^{15}N. What fraction of the resulting DNA molecules contain any ^{15}N?

A. 1/2
B. 0
C. 3/4
D. 1
E. 1/4

12. Which molecule regulates progression through the cell cycle by binding to cyclin-dependent kinases?

A. Tubulin
B. Histones
C. Cyclins
D. Collagen
E. Keratin

13. Per molecule of glucose, how many molecules of CO_2 are released in the complete aerobic oxidation (link reaction + Krebs cycle)?

A. 4
B. 6
C. 3
D. 12
E. 2

14. In humans, brown eyes (B) are dominant to blue eyes (b). Two brown-eyed parents have a blue-eyed child. What is the probability that their next child has brown eyes?

A. 1/2
B. 3/4
C. 2/3
D. 1
E. 1/4

15. What is the main role of the enzyme RuBisCO?

A. Splitting water in photosystem II
B. Synthesising ATP
C. Reducing $NADP^+$
D. Fixing CO_2 to ribulose bisphosphate
E. Producing oxygen

16. Which of the following increases the affinity of haemoglobin for oxygen (shifts the dissociation curve to the left)?

A. Increased CO_2
B. Decreased pH
C. Decreased temperature
D. Increased 2,3-BPG
E. Increased temperature

17. Which of the following is an unsaturated fatty acid characteristic?

A. One or more C=C double bonds
B. Contains no carboxyl group
C. Only single C–C bonds
D. Made of glucose units
E. Solid at room temperature in all cases

18. Which neurotransmitter is released at the neuromuscular junction of skeletal muscle?

A. Noradrenaline
B. Serotonin
C. GABA
D. Dopamine
E. Acetylcholine

19. Cardiac output equals:

A. blood pressure × heart rate.
B. heart rate + stroke volume.
C. heart rate × stroke volume.
D. end-diastolic volume − heart rate.
E. stroke volume / heart rate.

20. Genes A and B are linked. A test cross of AaBb (AB/ab) × aabb gives: AB 42%, ab 42%, Ab 8%, aB 8%. What is the recombination frequency?

A. 8%
B. 84%
C. 50%
D. 16%
E. 42%

21. Insulin lowers blood glucose mainly by:

A. stimulating gluconeogenesis.
B. stimulating glycogen breakdown.
C. increasing glucose uptake into muscle and adipose cells.
D. inhibiting glucose uptake into cells.
E. increasing glucagon secretion.

22. Which statement about DNA is correct?

A. A–T pairs have three hydrogen bonds.
B. The sugar is ribose.
C. Purines pair with purines.
D. The two strands run parallel.
E. G–C pairs have three hydrogen bonds.

23. Which of the following is part of the innate (non-specific) immune response?
A. Phagocytosis by neutrophils
B. Vaccination
C. Antibody production by plasma cells
D. Memory B cells
E. Clonal selection of T cells

24. Which substance is normally found in the glomerular filtrate?
A. Platelets
B. Large plasma proteins such as albumin in large amounts
C. Glucose
D. White blood cells
E. Red blood cells

25. Non-disjunction during meiosis I in a human produces gametes with chromosome numbers of:
A. 21 and 25
B. 23 and 23
C. 23 and 46
D. 22 and 24
E. 46 and 46

26. Denaturation of a protein by heat disrupts mainly which levels of structure?
A. Primary only
B. Peptide bonds
C. Secondary and tertiary
D. None
E. The amino acid sequence

27. Which of the following genetic disorders is inherited as autosomal dominant?
A. Huntington's disease
B. Haemophilia A
C. Phenylketonuria
D. Cystic fibrosis
E. Duchenne muscular dystrophy

28. The Na^+/K^+-ATPase pump moves:
A. 3 Na^+ out and 2 K^+ in per ATP.
B. 2 Na^+ out and 3 K^+ in per ATP.
C. 3 Na^+ in and 2 K^+ out per ATP.
D. Na^+ and K^+ in the same direction.
E. ions without using ATP.

29. Which antibiotic mechanism targets a structure absent in human cells?
A. Inhibition of the Golgi apparatus
B. Inhibition of the nucleus
C. Inhibition of peptidoglycan cell wall synthesis
D. Destruction of cholesterol in membranes
E. Inhibition of mitochondria

30. Which vitamin is absorbed in the terminal ileum and requires intrinsic factor?
A. Vitamin C
B. Vitamin A
C. Vitamin K
D. Vitamin B12
E. Vitamin D

31. The anticodon on a tRNA is 3'-UAC-5'. Which mRNA codon does it pair with?
A. 5'-GUA-3'
B. 5'-CAU-3'
C. 5'-ATG-3'
D. 5'-AUG-3'
E. 5'-UAC-3'

32. Genetic drift has the greatest effect on allele frequencies in:
A. small populations.
B. populations in Hardy–Weinberg equilibrium.
C. asexually reproducing bacteria only.
D. very large populations.
E. populations with high gene flow.

Section 4 — Chemistry (Q33–47)

33. What is the empirical formula of a compound with 40% C, 6.7% H and 53.3% O by mass? (C = 12, H = 1, O = 16)
A. CH_2O
B. C_2H_4O
C. CH_4O
D. CHO
E. $C_6H_{12}O_6$

34. Balance in acidic solution: $MnO_4^- + Fe^{2+} \rightarrow Mn^{2+} + Fe^{3+}$. How many Fe^{2+} ions react with one MnO_4^-?
A. 8
B. 5
C. 3
D. 1
E. 2

35. What is the pH of a solution formed by diluting 10 mL of 0.1 mol/L HCl to 1 L with water?
A. 5
B. 1
C. 2
D. 3
E. 4

36. Which element has the electron configuration [Ar] $4s^2$ $3d^6$?

A. Ni
B. Fe
C. Co
D. Mn
E. Cr

37. Which molecule has a bond angle of about 104.5°?

A. CH_4
B. BF_3
C. CO_2
D. H_2O
E. NH_3

38. In the reaction $N_2 + 3H_2 \rightarrow 2NH_3$, 28 g of N_2 reacts with 3 g of H_2. Which is the limiting reagent, and what mass of NH_3 forms? (N = 14, H = 1)

A. H_2; 17 g
B. Neither; 31 g
C. N_2; 17 g
D. H_2; 34 g
E. N_2; 34 g

39. Which of the following is a ketone?

A. Propanone
B. Propanoic acid
C. Propanal
D. Methyl ethanoate
E. Propan-1-ol

40. What is the osmolarity of a 0.15 mol/L NaCl solution (assuming complete dissociation)?

A. 0.60 osmol/L
B. 0.075 osmol/L
C. 0.30 osmol/L
D. 0.15 osmol/L
E. 0.45 osmol/L

41. Which of the following substances conducts electricity when solid?

A. Diamond
B. Ice
C. NaCl
D. Sugar
E. Graphite

42. The Ka of a weak acid HA is 10^{-5}. What is the pH of a buffer containing equal concentrations of HA and A^-?

A. 9
B. 3
C. 7
D. 5
E. 10

43. Which statement about a reaction with $\Delta H < 0$ is correct?

A. It is always spontaneous.
B. It releases heat to the surroundings.
C. It absorbs heat from the surroundings.
D. It never needs activation energy.
E. Its products have more enthalpy than reactants.

44. How many structural isomers are there for C_5H_{12}?

A. 6
B. 5
C. 4
D. 2
E. 3

45. Which of the following pairs are isotopes?

A. ^{40}K and ^{40}Ca
B. H_2O and D_2O molecules
C. Diamond and graphite
D. O_2 and O_3
E. ^{12}C and ^{14}C

46. Given $E°(Cu^{2+}/Cu) = +0.34$ V and $E°(Zn^{2+}/Zn) = -0.76$ V, what is the standard emf of a Zn/Cu cell?

A. 0.34 V
B. −1.10 V
C. 0.76 V
D. 1.10 V
E. 0.42 V

47. Which type of reaction occurs when an alkane reacts with chlorine under UV light?

A. Nucleophilic substitution
B. Free-radical substitution
C. Condensation
D. Elimination
E. Electrophilic addition

Section 5 — Physics & Mathematics (Q48–60)

48. A projectile is launched horizontally from a cliff 45 m high at 10 m/s. How far from the base of the cliff does it land? ($g = 10$ m/s^2, no air resistance)

A. 10 m
B. 30 m
C. 20 m
D. 45 m
E. 90 m

49. A 5 kg block slides on a horizontal surface with coefficient of kinetic friction 0.2. What is the friction force? ($g = 10\ m/s^2$)

A. 1 N
B. 25 N
C. 5 N
D. 10 N
E. 2 N

50. A 10 µF capacitor is charged to 20 V. What charge is stored?

A. 2 µC
B. 0.5 µC
C. 30 µC
D. 2000 µC
E. 200 µC

51. Light passes from air (n = 1.0) into glass (n = 1.5). Its speed in glass is: ($c = 3 \times 10^8$ m/s)

A. 4.5×10^7 m/s
B. 3×10^8 m/s
C. 2×10^8 m/s
D. 1.5×10^8 m/s
E. 4.5×10^8 m/s

52. Blood flows through an artery of cross-sectional area 2 cm^2 at 30 cm/s. It then flows into a narrowed section of area 0.5 cm^2. What is the speed there?

A. 120 cm/s
B. 240 cm/s
C. 15 cm/s
D. 7.5 cm/s
E. 60 cm/s

53. A 0.5 kg ball hits a wall at 8 m/s and rebounds at 6 m/s. What is the magnitude of the change in momentum?

A. 14 kg·m/s
B. 3 kg·m/s
C. 7 kg·m/s
D. 4 kg·m/s
E. 1 kg·m/s

54. An ideal heat engine operates between 600 K and 300 K. What is its maximum (Carnot) efficiency?

A. 100%
B. 67%
C. 50%
D. 25%
E. 33%

55. If $\log_3(x) = 4$, what is $\log_9(x)$?

A. 1
B. 8
C. 16
D. 4
E. 2

56. What is the period of the function $y = \sin(2x)$?

A. $\pi/2$
B. 4π
C. 1/2
D. 2π
E. π

57. In a group of 10 students, how many different committees of 3 students can be formed?

A. 720
B. 30
C. 60
D. 1000
E. 120

58. What is the centre of the circle $x^2 + y^2 - 4x + 6y - 3 = 0$?

A. $(-2, 3)$
B. $(2, -3)$
C. $(-4, 6)$
D. $(4, -6)$
E. $(2, 3)$

59. What is the sum of the first 10 terms of the arithmetic sequence 3, 7, 11, ...?

A. 400
B. 210
C. 230
D. 190
E. 39

60. A charge of 2 μC is placed in a uniform electric field of 5000 N/C. What force acts on it?

A. 0.1 N
B. 0.01 N
C. 2500 N
D. 10 000 N
E. 0.001 N

정답 및 해설

번호	정답	해설
1	C	영상 연구는 위약 반응에 생물학적 근거(내인성 오피오이드)가 있음을 보여 주기 위함.
2	B	위약으로 치료할 수 없고 효과적 치료가 따로 있는 질환이 있어야 "대체는 비윤리적"이 성립.
3	E	알렉산더 플레밍.
4	B	노르웨이는 EEA 회원국이지만 EU 비회원.
5	A	후건 긍정의 오류 → 비가 왔다고 확정 불가.
6	D	12/80 = 15%.
7	A	Maria=화학, Luca는 생물·물리 아님 → 해부학. Nina는 물리 아님 → 생물. Omar=물리.
8	E	모서리 중간 큐브: 12개 모서리 × 1 = 12.
9	D	5×20 = 100 mg → 100/25 = 4 mL.
10	E	내공생설 근거: 원형 DNA, 70S 유사 리보솜, 이중막.
11	A	2회 복제 후 4분자 중 2분자가 ^{15}N 가닥 포함 → 1/2.
12	C	사이클린-CDK 복합체.
13	B	연결반응 2 + 크렙스 4 = 6 CO_2.
14	B	부모 Bb × Bb → 갈색 3/4.
15	D	CO_2를 RuBP에 고정.
16	C	온도↓ → 친화력↑(좌측 이동). 나머지는 우측 이동.
17	A	C=C 이중결합 존재.
18	E	아세틸콜린.
19	C	CO = HR × SV.
20	D	재조합형 8+8 = 16%.
21	C	근육·지방세포 GLUT4 → 포도당 흡수↑.
22	E	G–C 수소결합 3개. 두 가닥은 역평행.
23	A	호중구 식세포작용(선천면역).
24	C	포도당은 여과됨(이후 재흡수).
25	D	감수 1분열 비분리 → 24와 22 (n+1, n−1).
26	C	2·3차 구조 파괴, 펩타이드 결합(1차)은 유지.
27	A	헌팅턴병.
28	A	3 Na^+ 밖, 2 K^+ 안.
29	C	펩티도글리칸 세포벽(β-락탐 등).
30	D	비타민 B12 + 내인자, 회장 말단.
31	D	안티코돈 3'-UAC-5' ↔ 코돈 5'-AUG-3'.
32	A	작은 집단.
33	A	C 3.33 : H 6.7 : O 3.33 → 1:2:1 → CH_2O.
34	B	Mn +7→+2 ($5e^-$), Fe +2→+3 ($1e^-$) → 5 Fe^{2+}.
35	D	10^{-3} mol in 1 L → pH 3.

번호	정답	해설
36	B	Fe(26).
37	D	H_2O 굽은형 104.5°.
38	A	N_2 1 mol은 H_2 3 mol 필요, H_2는 1.5 mol뿐 → H_2 한정, NH_3 1 mol = 17 g.
39	A	프로파논(아세톤).
40	C	$Na^+ + Cl^- \rightarrow 2 \times 0.15 = 0.30$.
41	E	흑연(비편재 전자).
42	D	헨더슨–하셀바흐: pH = pKa = 5.
43	B	발열 반응. 자발성은 ΔG로 결정.
44	E	n-펜테인, 2-메틸뷰테인, 2,2-다이메틸프로페인.
45	E	양성자 수 같고 중성자 수 다름.
46	D	$0.34 - (-0.76) = 1.10$ V.
47	B	자유 라디칼 치환.
48	B	낙하 시간 $\sqrt{(2 \cdot 45/10)} = 3$ s → $10 \times 3 = 30$ m.
49	D	$\mu mg = 0.2 \times 5 \times 10 = 10$ N.
50	E	$Q = CV = 10\ \mu F \times 20$ V $= 200\ \mu C$.
51	C	$v = c/n = 2 \times 10^8$.
52	A	$A_1v_1 = A_2v_2 \rightarrow 2 \times 30/0.5 = 120$ cm/s.
53	C	방향 반전: $0.5 \times (8+6) = 7$.
54	C	$1 - 300/600 = 50\%$.
55	E	$x = 81 = 9^2 \rightarrow 2$.
56	E	$2\pi/2 = \pi$.
57	E	${}_{10}C_3 = 120$.
58	B	$(x-2)^2 + (y+3)^2 = 16$ → 중심 $(2, -3)$.
59	B	$a=3, d=4$: $10/2 \times (2 \cdot 3 + 9 \cdot 4) = 5 \times 42 = 210$.
60	B	$F = qE = 2 \times 10^{-6} \times 5000 = 0.01$ N.